Techno-economical Technology Assessment for Operational Zero Carbon Supermarkets

Salvador Acha[a], Georgios Mavromatidis[b], Victor Caritte[c] and Nilay Shah[d]

[a] *Imperial College London, London, UK, salvador.acha@ic.ac.uk*
[b] *Imperial College London, London, UK, georgios.mavromatidis11@ic.ac.uk*
[c] *Imperial College London, London, UK, victor.caritte11@ic.ac.uk*
[d] *Imperial College London, London, UK, n.shah@ic.ac.uk*

Abstract:
Supermarkets are very energy intensive buildings resulting in a large carbon footprint that needs to be addressed. Demand reduction measures are not sufficient, since even after intensive actions have been taken, the remaining footprint is still significant. Therefore, measures that focus on the energy supply side are necessary in order to minimize or preferably eliminate carbon emissions. In their quest towards commercial sustainability, food retail businesses require relevant frameworks that can help them identify the various pathways in which they can reduce operational carbon emissions. These frameworks need to be straightforward as decision makers generally lack the engineering expertise such projects demand. This paper focuses on depicting the various challenges low carbon technologies need to overcome to be successfully applied in supermarkets while also identifying core technologies that can deliver sustainable supermarkets today; a review of multiple technologies is presented and graded according to a strict sustainability criteria. A generic UK supermarket is used to describe the different store services and to characterize daily, seasonal, and annual energy load profiles in the food retail industry. These are used to explain the complex relationship between electricity and heat requirements the building has throughout a year. A business as usual store design is presented to benchmark current key performance indicators (KPIs) concerning emissions, capital and operational costs; these KPIs are used by decision makers when approving a future sustainable building design. After a benchmark has been established, it possible to assess and compare various low carbon energy system designs to identify the trade-offs each alternative offers. Results of the analysis indicate designing and delivering a zero carbon supermarket is possible, however careful detail must be dedicated to address multiple issues that come along with project implementation; such as the manner in which the energy systems should be operated or potential hazards to customers or employees. Useful guidelines on how energy technologies can complement each other are included. Once the list of low carbon technologies has been completed for a supermarket environment, the paper expands by stating issues to be on the look-out concerning project implementation and operational life time of the energy system. Overall, defining the technologies going into a sustainable building is just one step towards achieving a zero carbon supermarket, nonetheless it is its most critical stage, where trade-offs are assessed and priorities are given to guarantee the success of such ambitious projects.

Keywords:
Zero carbon buildings, Low carbon technologies, CHP, Supermarkets, Sustainability.

1. Introduction

According to IEA, on a global scale, residential and commercial buildings consume approximately 32% of global energy use and are responsible for 10% of direct CO2 emissions and over 30% of total end-use energy-related emissions. Moreover, energy demand in the built environment is projected to double by 2050, mainly driven by the increasing number of new residential and commercial buildings due to expanding global population [1].

Amongst the different energy saving options that are available for the built environment, the adoption of the Zero Energy Building (ZEB) concept has the greatest potential for the mitigation of CO2 emissions and the reduction of energy use. According to Marszal et al [2], the ZEB concept should not be seen as a distant goal since it can be realized with today's building construction means and available energy system technologies; as it can be attested via live demonstration projects and academic research activities [3-4].

In the United States, the Energy Independence and Security Act of 2007 [5] established the Zero-Net-Energy Commercial Buildings Initiative with the goal to deliver a zero net energy commercial building for new builds by 2030. But it does not stop there and it shows ambition by also aiming to impact 50% of the commercial building stock of the US by 2040 and finally all commercial buildings by 2050. In Europe, the European Performance of Buildings Directive and its recast [6] require that in all Member States new buildings occupied and owned by public authorities are nearly zero-energy buildings while by 31 December 2020 all new buildings should be nearly zero-energy buildings.

From the different types of commercial buildings, supermarkets today are very energy intensive in their operation due to their high use of refrigeration. In the United States (US) and France, the electricity consumption by supermarkets is equal to 4% of total national electricity use [7]. In the UK, supermarkets consume approximately 12 TWh, which is equal to 3.5% of the total UK electric energy consumption [8-9]. Moreover, the CO2 emissions that can be attributed to energy consumption are approximately 4 MtCO2 which corresponds to 1% of UK's greenhouse gas emissions [9-11].

Even though considerable efforts have taken place to reduce energy consumption of supermarkets, the remaining footprint is still significant [12]. Therefore, effective application of the ZEB scheme seems to be the logical way forward for food retailers.

The purpose of this paper is to provide a high-level framework and a review of technologies for decision makers in order to assist in the design of zero carbon supermarkets. Section 2 reviews different definitions of a ZEB and concludes with the development of the zero carbon supermarket definition. Different energy efficiency options that are available for supermarkets before the application of a ZEB concept are outlined in Section 3. Section 4 reviews different technologies for the provision of heat and electricity to a supermarket that comply with the ZEB definition, as well as of complementary technologies that can enhance performance. Section 5covers the idiosyncrasies of heat and electricity demand in supermarket buildings, the performance indicators used to assess a net zero carbon building and shows the trade-offs for different alternative solutions. Section 6 concludes by reviewing the analysis performed and by indicating the different issues surrounding a successful implementation of a net zero commercial carbon building.

2. Zero energy building definitions

No exact definition can be drawn upon among the different publications describing and evaluating different Zero Energy Building (ZEB) designs. Although there is increasing global interest in the ZEB concept, this lack of common understanding implies it is difficult to narrow the scope and adequately define what is to be addressed in this complex endeavour. For instance, one source of misunderstanding arises from the metric that should be used to define zero. Different authors have conflicting criteria since they have employed energy (end-use or primary), exergy, CO2 emissions or energy costs in as their key objective in their analysis [13].

Torcellini and Crawley [14] investigated these issues and developed different definitions for different ZEB cases: net zero source energy building, net zero site energy building, net zero energy cost building, and net zero energy emissions building; these are:

- A *net zero source energy building* produces as much electricity as it uses in a year in source energy terms. This refers to primary energy used to generate and deliver energy to the site.
- A *net zero site energy building* produces the same amount of energy as it uses in a year, when measured at the site.

- A *net zero energy cost building* is achieved when the money paid by the utility to the building owner for energy export is equal to the cost of energy imported from the grid. This can be difficult to achieve sometimes due to usually very low utility rate tariffs.
- A *net zero energy emissions building* use as much emissions from renewable energy as it would from conventional emissions-producing sources.

Marszal and Heiselberg [13] suggested a list of topics that should be considered before developing a new ZEB definition; these are:

- What is the metric that will be used to fulfil the criteria? (e.g. energy costs; CO2 emissions)
- Which energy demands are to be considered in the assessment?
- Should the embodied energy of the building materials and technical equipment used be considered in the balance?
- Finally, should specific requirements be included with regards to maximum energy production and use? Should indoor air quality (IAQ) and type of renewable energy sources be considered as well?

For the purpose of this paper, the authors have used the aforementioned guidelines to develop a definition for a net zero carbon supermarket (NZCS). The word net has been adopted to illustrate, as is mentioned in [13], that there is interaction with the utility grid. The metric that is used is the CO2 emissions on a yearly basis and heat and electricity use are considered for the balance, however the embodied energy has been discarded. Finally, no specific requirements exist for the IAQ or the type of energy sources used. Overall, in order to create a zero carbon supermarket the following equation needs to be correct.

$$(On - site\ energy\ emissions) + (Elec\ imports - Elec\ exports) \times Grid\ Carbon\ Factor = 0 \qquad (1)$$

Therefore, if zero carbon renewable sources are used on site, simply the electricity exports need to be equal to imports. On the other hand, if a low carbon fuel is used on-site then exports to the grid need to be larger than imports to account for the carbon emissions on-site.

However, even if a uniform NZCS definition is obtained, unlike passive house standards, there are no exact guidelines on how to deliver the technology solution for a zero energy building [15]. As Marszal and Heiselberg [13] state, the majority of publications that describe the road to ZEB start with the consideration of energy efficiency measures in order to minimise energy demand and afterwards utilise renewable or low carbon energy sources to supply the remaining energy. Thus, it is at energy reduction measures, better known as energy efficiency strategies, where we begin our journey towards decarbonisation.

3. Decarbonisation supermarket journey

The majority of the energy conservation measures that food retailers take focus mainly on the energy demand side trying to curb energy consumption of the main energy consumers in the store, namely Refrigeration, Lighting and HVAC (Heating, Ventilation & Air Conditioning) systems.

With regards to supermarket refrigeration systems, several options exist to drastically decrease the energy consumption. The most prevalent of them include: installing glass doors on refrigerated cabinets [16], use of alternative refrigerants [17], indirect/secondary loop systems [18], mechanical subcooling [19], floating condensing [20], heat recovery systems [20], cabinet LED lighting [9] and defrost-on-demand schemes [21]. A very comprehensive review of all the energy efficiency options for commercial refrigeration systems can be found in [22].

For the lighting system, the most commonly used approaches to curtail energy consumption include the use of LED lighting [10], use of natural daylight in conjunction with digital dimming system [23] and finally occupancy sensors [24].

Finally, as regards HVAC systems, apart from using more efficient system components, such as condensing boilers or higher COP electric chillers, other energy efficiency measures include positive pressurization of the store to minimize leakage of the unconditioned and unfiltered air from the outside [25], Use of variable speed drives (VSDs) for air handling units (AHUs) [26] and integration of the HVAC system with the store's refrigeration system [27-28].

Even if all energy conservation measures that were analyzed in the literature are implemented in a store, energy demand cannot be eliminated completely and if energy is supplied by conventional sources then the carbon footprint will still be significant. To illustrate that point, a statistical analysis on energy intensity of 2,570 UK supermarkets conducted by Tassou et al [9], illustrated in Fig. 1, shows that almost no stores are below 500 kWh/m2 (~46 kWh/ft2). Therefore, it becomes evident that measures focusing on the energy supply side are necessary to decarbonize the energy demand of stores – no matter its size, shape, or location.

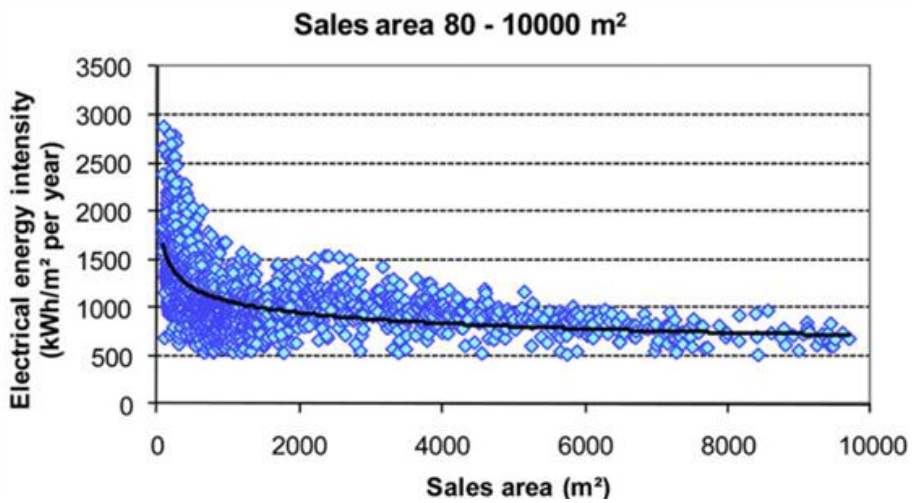

Fig. 1. Variation of electrical energy intensity of 2570 UK retail food stores with sales area from 80 m^2 to 10,000 m^2 [9].

On the supply side, an initial step taken by many food retailers is to replace natural gas for use in space heating applications with biomass/biofuel boilers. This could effectively decarbonize thermal energy supply. However, space heating related emissions in UK supermarkets account only for 10% to 20% of the total energy related emissions [10]; hence the remaining footprint is still significant. Similarly, installation of PV panels on the roofs of supermarkets can contribute towards cleaner electricity use on-site; nevertheless, space constraints and generation limitations do not allow enough capacity to be installed to effectively decarbonize all store operations while having the drawback of not producing energy at night time.

Finally, cogeneration or combined heat and power (CHP) seems like a viable option for energy savings and efficiency. However, the demand for space heating is greater during the winter months and in the summer a considerable quantity of heat needs to be rejected in the atmosphere, since the seasonal profile for the heat and electricity demand do not match. Therefore, to achieve high plant utilization the system could be coupled with a thermally-driven absorption chiller that can assist refrigeration systems or provide space cooling (trigeneration system). Several studies have investigated the potential of trigeneration schemes in supermarkets [29-34]. However, if a fossil fuel source is used for the prime mover even though energy savings and emissions curtailing can be achieved, it is still far from zero carbon.

4. Technology review

4.1. Electricity Generation

Electricity accounts for approximately 70% of the energy consumption in a supermarket in the UK [10]. In regions with milder climates with less space heating but more space cooling demand this ratio would be even higher. Therefore, it is only logical to focus on electricity provision technologies first. The first division is for technologies that can be classified as dispatchable and non-dispatchable. The term dispatchable refers to a technology that can generate electricity on demand; these technologies are usually used to provide the base load of the building. Non-dispatchable technologies are ones that are intermittent and energy can only be utilized when it is generated unless a storage system is used.

In order to evaluate the applicability of each technology for a zero carbon supermarket, a "Red-Amber-Green" scheme is utilized. The green label indicates that the technology fulfils zero carbon store objectives; the amber that the technology has some issues that need addressing for implementation and finally the red means the technology does not meet the requirements of the project. Five categories were used in order to get a better understanding on different issues of each technology and were graded using the aforementioned scheme; these are:

- *Reliability* is used to indicate whether or not the technology is prone to break downs or has any complicated maintenance requirements;
- *Emissions* refers to CO2 emitted during the system's operation;
- *Capacity* indicates whether the available range of capacities is close to the store peak load magnitude;
- *Suitability* refers to installation suitability for a commercial environment;
- *Maturity* is used to indicate the track-record of the technology with regards to market availability, number of installations etc.

In terms of dispatchable sources, hydroelectric and geothermal technologies can be utilized since they are mature and reliable technologies and are effectively zero carbon. However, the installed capacity of such schemes is much higher than what store needs, thus the economy of scales do not translate. Moreover, suitability issues arising from the availability of water sources and geothermal fields close to the store make these technologies relatively difficult to choose. Cogeneration or Combined Heat and Power (CHP) schemes can provide the store with both electricity and heat with increased efficiencies and utilising a wide range of fuels. For the objective of a zero carbon store, two available options have been identified: a bioenergy CHP and a fuel cell CHP.

With regards to the different bioenergy options for the CHP engine, these can be solid, liquid or gaseous fuels. Beginning with solid fuels the most established options include wood chips and wood pellets. Issues with that fuel type include the need for additional pieces of equipment such as a gasifier to allow the utilization of the fuel in a CHP engine or alternatively a biomass boiler to raise steam that can then expand in a steam turbine. Both these configurations though increase complexity, maintenance costs and decrease efficiency due to parasitic load for the operation of the extra kits. In terms of liquid biofuels the most prevalent options include bioethanol and biodiesel. Both fuels have an established supply chain and can be considered future-proof. However, bioethanol's volatility issue makes it more hazardous in commercial environment compared to biodiesel, which seems to be a more attractive option. In the UK, both solid and liquid fuels are supported with incentives (Renewable Obligation Certificates – ROCs) [35]. Finally, for the gaseous fuel, the only option is biogas via an Anaerobic Digestion plant or using landfill gas. Using biogas has the advantage that it can be easily utilized in a modified natural gas engine; however supply chain not being in place in conjunction with storage complications together with the opaque policy associated with biogas in the UK hinders its adoption [36].

Using a fuel cell as a CHP engine means that there are two fuel options: hydrogen or natural gas. Even though hydrogen would offer zero carbon emissions, there is still no infrastructure in place for its supply. Using natural gas instead, even if there are lower emissions compared to an IC engine CHP due to higher electrical efficiency it still cannot contribute effectively towards a zero carbon store. Moreover, fuel cells can still be considered an immature technology with several issues like reliability and commercial suitability needing to be resolved [37].

In terms of non-dispatchable sources, wind turbines are a mature technology that can provide the store with clean electricity. Certain issues exist though and they include intermittency and capacity issues. The latter refers to the generating capacity of the turbines being larger than the demands of the store. But most importantly, the issue with wind turbines is the location. Stores are sometimes built in urban areas which are not suitable for the installation of wind turbines. On the other hand, if buildings cannot be zero carbon they should consider the installation of wind turbines at remote sites to supply a number of stores with energy; usually known as carbon off-setting [38].

Finally, utilising the solar resource via solar PV or hybrid PV is a viable option. Unlike the first option that fulfils all the criteria but can only generate electricity, the latter manages to capture some of the heat that is dissipated by the panel and use it for hot water purposes, for instance. However, there are still issues with capacity, reliability and most importantly maturity [39].

Table 1. Technology evaluation for electricity generation.

Dispatchable Generation Technologies					
Technology	*Reliability*	*Emissions*	*Capacity*	*Suitable*	*Mature*
Hydro	🟢	🟢	🟡	🔴	🟢
Geothermal	🟢	🟢	🟡	🟡	🟢
Bioenergy CHP	🟢	🟢	🟢	🟢	🟡
Fuel Cell CHP	🟡	🟡	🟢	🟡	🟡
Non-dispatchable Generation Technologies					
Technology	*Reliability*	*Emissions*	*Capacity*	*Suitable*	*Mature*
Wind turbines	🟡	🟢	🟡	🔴	🟢
Solar PV	🟢	🟢	🟢	🟢	🟢
Hybrid PV	🟡	🟢	🟡	🟢	🔴

4.2. Heat Generation

In the supermarket context, heat is needed for space heating and hot water services. Similarly to electricity generation options, the “Red Amber Green” scheme is used for evaluation.

Boiler technologies are the first type of appliances considered for the supply of heat to the store. Firstly, an electric boiler can be utilized with a few minor issues only in terms of capacity. The emissions indicator has all three grades depending on the carbon intensity of the electricity source [40]. Alternatively, a bioenergy driven boiler, utilising solid, liquid or gaseous fuel can be used. The only issue identified by the authors is the maturity of the technology with the same conclusions for the different types of fuel being applicable in this case too.

Heat pump technology can also be employed to provide the store with thermal energy [41]. Either Air-source heat pumps (ASHP) or Ground-source Heat Pumps can be installed with the first one being less efficient but more mature than the latter. Both options have been given a green/amber point for the emissions indicator since emissions depend on the source of electricity – conventional source/grid or low carbon solution.

Finally, non-dispatchable generation, options include either solar thermal technologies for the provision of thermal power only or hybrid PV technologies that can provide electricity similarly to a normal PV panel however some heat can be recovered via a panel cooling circuit [42]. With regards to the criteria set to evaluate the different technologies, the first system meets all requirements; however, the latter faces reliability and capacity issues arising mainly from the fact that it is still a fairly immature technology.

Table 2. Technology evaluation for heat generation.

Dispatchable Generation Technologies					
Technology	*Reliability*	*Emissions*	*Capacity*	*Suitable*	*Mature*
Hydro	🟢	🟢	🟡	🔴	🟢
Geothermal	🟢	🟢	🟡	🟡	🟢
Bioenergy CHP	🟢	🟢	🟢	🟢	🟡
Fuel Cell CHP	🟡	🟡	🟢	🟡	🟡
Non-dispatchable Generation Technologies					
Technology	*Reliability*	*Emissions*	*Capacity*	*Suitable*	*Mature*
Wind turbines	🟡	🟢	🟡	🔴	🟢
Solar PV	🟢	🟢	🟢	🟢	🟢
Hybrid PV	🟡	🟢	🟡	🟢	🔴

4.3. Complementary technologies

Apart from the main energy conversion technologies outlined above, several complementary technological options exist that can increase the efficiency of the installation and/or enhance the operation flexibility.

The first option includes the utilization of an absorption chiller that can be driven with heat generated by a CHP or a solar thermal and is not needed for space heating/hot water applications. By utilising heat that would otherwise be discarded, the overall efficiency of the system is increased. The main options for absorbent-refrigerant pairs for absorption chillers are LiBr – H2O and H2O-NH3.(i.e. ammonia). However, due to safety implications associated with NH3, the only option is the first pair. Since H2O is the refrigerant in this case, the lower temperature that can be achieved will be above 0°C, hence the system can be utilized for either space cooling applications or to assist the refrigeration system via subcooling of the refrigerant or via a water-cooled condensing scheme [43].

Alternatively, another way to recover waste heat is Organic Rankine Cycle (ORC) systems. An ORC engine uses an organic fluid with low boiling point instead of water that is favorable for low temperature applications. Actual applications include recovering of geothermal heat or waste heat from power generation [44]. Its advantages include great flexibility, high safety and low maintenance requirements [45]. The use of an ORC in a zero carbon supermarket can be used to recover any waste heat generated and increase the overall electrical efficiency of the installed systems.

Finally, thermal storage either for heat or cooling can be utilized to allow operating flexibility for the installed systems by allowing energy to be stored during periods of low demand for use later during peak demand times. Several options exist and can be either sensible thermal storage, using liquid (typically water) or solid materials (soil and stone), or latent storage using Phase Change Materials (PCMs) [46].

5. Determining a net zero carbon solution

5.1. Heat to electricity ratios in supermarkets

In order to achieve zero carbon operation in stores it is necessary to generate low carbon electricity on-site. PV panels can help toward this issue but the amount of panels stores can get is limited depending of the size of the store. On an annual basis, the resulting electricity generation is not sufficient to cover stores' energy demand. Therefore, stores would need to employ a generator running on a low carbon fuel. Additionally, wasted heat could be used as part of a CHP scheme for stores' space heating and hot water usage.

Preliminary calculations showed that it would not be possible to use all the heat rejected by the CHP as a supermarket uses more electricity than heat on an annual basis as illustrated on Fig. 2a and 2b. As days get warmer in summer, refrigeration system electrical load increases resulting in a higher supermarket electricity demand. Conversely, heat demand for space heating decreases drastically in summer. This phenomena changes substantially in cold weather spells, since heat demand overwhelms electricity demand. This heat to power (HTP) ratio variability throughout the year creates the necessity for on-site energy solutions in supermarkets to be very flexible in the manner they are managed.

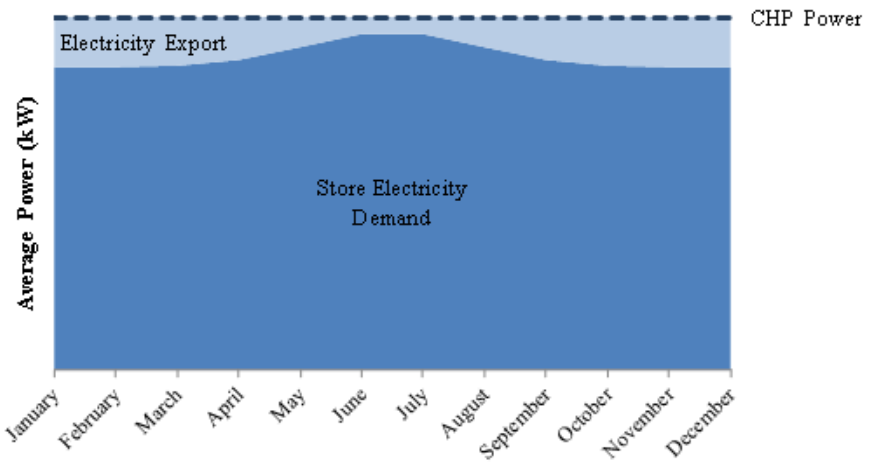

Fig. 2a. Typical supermarket electricity profile with CHP production.

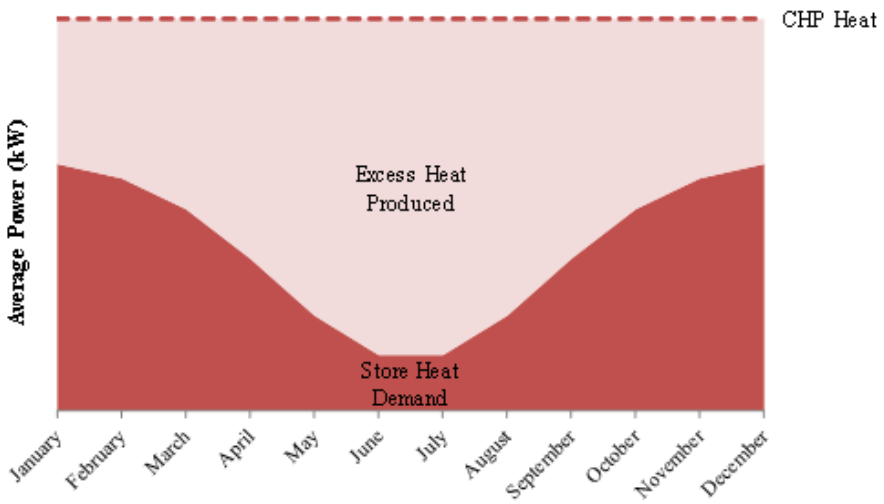

Fig. 2b. Typical supermarket heat profile with CHP production.

On an annual basis, a load duration curve as seen in Fig. 3 depicts that electricity demand is indeed significantly higher than heat demand throughout the year, except for 450 hours in the coldest periods of the day (5% of the time). To mitigate the waste of excess heat from the CHP unit, technologies such as ORC and absorption chillers (as detailed in Section 4.2) could be used.

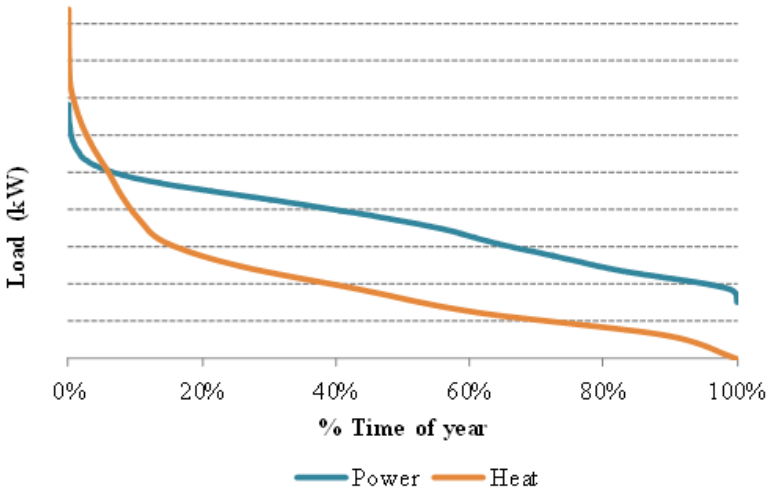

Fig. 3. Power and electricity load duration curves.

5.2. Evaluation KPIs

It is particularly important to set a robust Business as usual (BAU) solution against which the different designs will be compared. The BAU solution can be described with the Fig. 4 schematic consisting of important gas and electricity to satisfy the energy needs of the store.

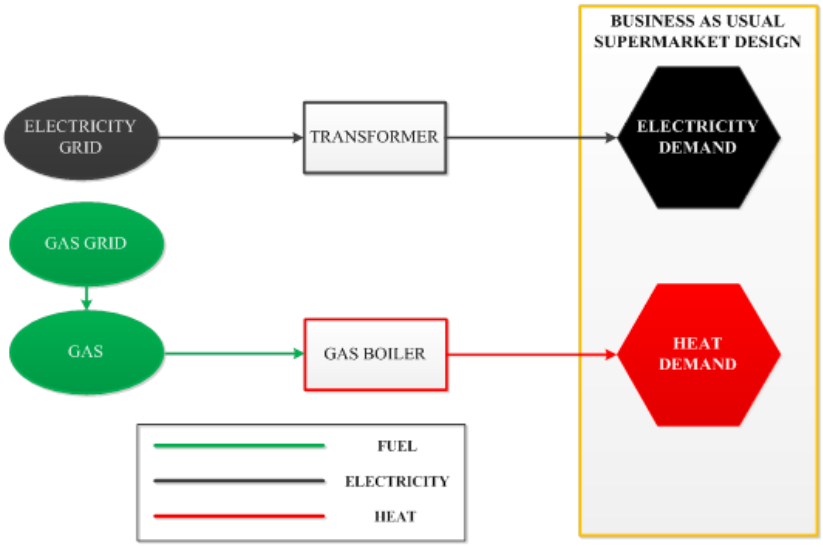

Fig. 4. Business as usual system layout.

After carefully reviewing the expected energy demands a new build will have and the set of technologies available to decarbonise its operations, a thorough exercise is needed to assess both the high-level and day to day performance of alternative solutions. Each design naturally will have its trade-offs not just on capital and operational expenditure, but also in the practicality of achieving net zero carbon. In this paper, the features of a new build 42,000 ft2 store in the South of England were taken to showcase the range of solutions. Such a building would benefit from state-of-the-art

building properties; such as adequate air tightness, good u-values, and the incorporation of sky lights. The energy requirements of the future supermarket are estimated and based on the actual performance of similar stores, normalised per square foot and adjusted for weather variation. By having a good forecast of the energy requirements of the building it is possible to benchmark emissions, CAPEX and OPEX for the BAU scenario.

Subsequently, the final decision is based on the following indicators of success. Firstly, the decarbonisation factor achieved. Based on the definition of a zero carbon building in section 2, shown by the following indicator:

$$Decarbonisation\ factor = \frac{BAU\ emissions - New\ net\ emissions}{BAU\ emissions} \tag{2}$$

When this factor reaches 100%, it means that the solution is zero carbon with respect to the definition we gave. Net new emissions are influenced by the GHG intensity of the fuel being used and by the efficiency of the engine in which it is consumed.

Secondly, CAPEX and OPEX were compared against Business as Usual. It seems obvious that the solution proposed will have a higher CAPEX than the BAU grid and gas boiler solution. However, if the business case shows that substantial savings in OPEX are possible; then the business case of the project can gain much momentum.

Thirdly, the share of heat wasted. This indicator will be calculated both in the context of a CHP and of a CCHP to help decision makers assess the need for an absorption chiller that could help diverting a substantial amount of wasted heat and thus help the scheme to qualify easily for Quality Assurance for CHPs in the UK (CHPQA) [47].

5.3. Assessing a net zero carbon store solution

The 42,000 ft2 used for the case study would have a total electricity demand of 1,830 MWh and heat demand of 1,070 MWh per annum. Fig. 5 details the expected energy pie of such a building; where heat and refrigeration almost consume two-thirds of all energy needs. The net zero carbon building design shown in Fig. 6 was proposed to see how it compares against BAU scenario. Since the new building would enjoy great thermal properties and thus not require a lot of heat demand, the design considers a CCHP scheme with the objective to maximise the use of excess heat by making the chiller have a high load factor. An ORC could have also been considered to further reduce the amount of heat wasted. However, preliminary calculations showed that in most cases, the absorption chiller would be sufficient to recover most of this heat waste and that the ORC would therefore be redundant.

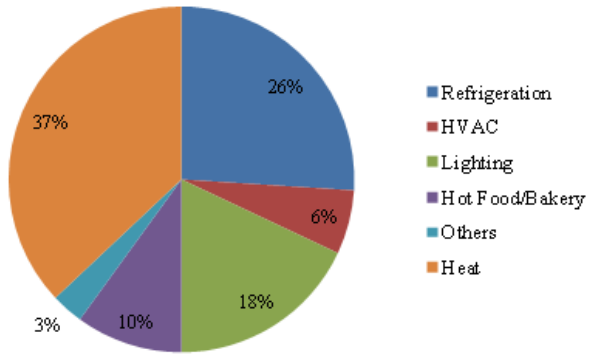

Fig. 5. Expected energy distribution of the new store.

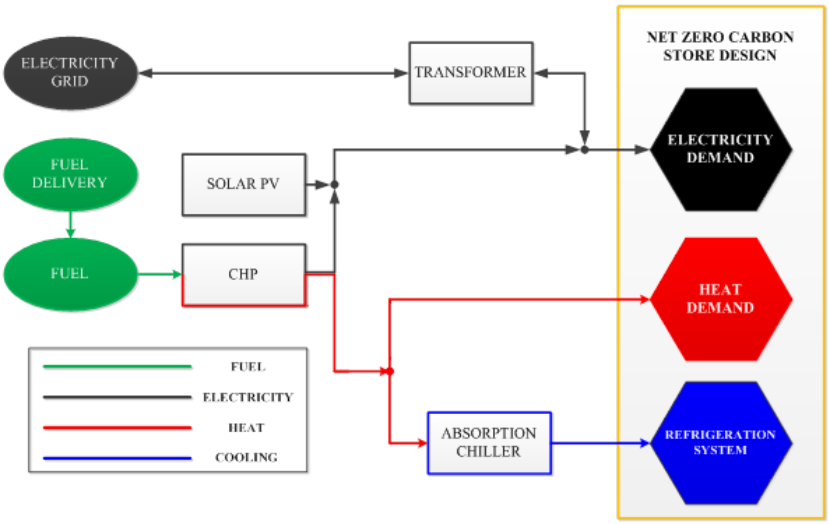

Fig. 6. Zero carbon stores proposed design layout.

Based on projected energy demand, the residual carbon number to be offset by the CHP scheme was estimated and the simulation was conducted considering 19 engines. These 19 engine alternatives are result of a limited combination of fuel and technology selected at the design stage; the following technologies were considered – see Table 3.

Table 3. Zero carbon stores technology considered.

Technology	*Fuel*	*Power capacity*	*Notes*
NG CHP	Natural Gas	300 kW to 1.2 MW	Reciprocating engine. Large range of capacities were considered since gas off-sets grid carbon by just a fraction in the UK (no incentive)
Biofuel CHP	Biodiesel	30 kW to 600 kW	Fuel considered was a triglyceride biofuel running on a modified diesel engine (incentive 2ROCs)
Biomass CHP + Gasifier	Wood Chips	60 kW to 240 kW	Gasified wood chip fuel coupled with a Stirling engine (incentive 2ROCs)
NG CHP + biogas injection	Natural Gas	300 kW to 600 kW	Same solution as NG CHP but off-setting carbon by purchasing equal volumes of AD biogas (no incentive)

5.4. Results

A techno-economical study was performed comparing how each alternative solution performed with the energy requirements of the proposed building throughout a year.

The two first indicators of success: decarbonisation and costs were plotted together. Fig. 7shows the capital expenditure on the y-axis and decarbonisation factor on the x-axis for the different solutions. The figure shows first of all that good decarbonisation progress is achievable, however in order to reach a net zero carbon building CAPEX becomes very high; particularly for biomass and natural gas solutions. Biofuel and biogas injection to the grid seem more sensible. Meanwhile, the evolution of operational expenditure against decarbonisation factor can be seen on Fig. 8. OPEX costs for some of the solutions are equal or are more competitive than BAU; however for all the cases except biofuel as engines get bigger the OPEX increases exponentially. This means that until fuel costs get cheaper zero carbon building will be very hard to achieve as the business case is simply not there yet.

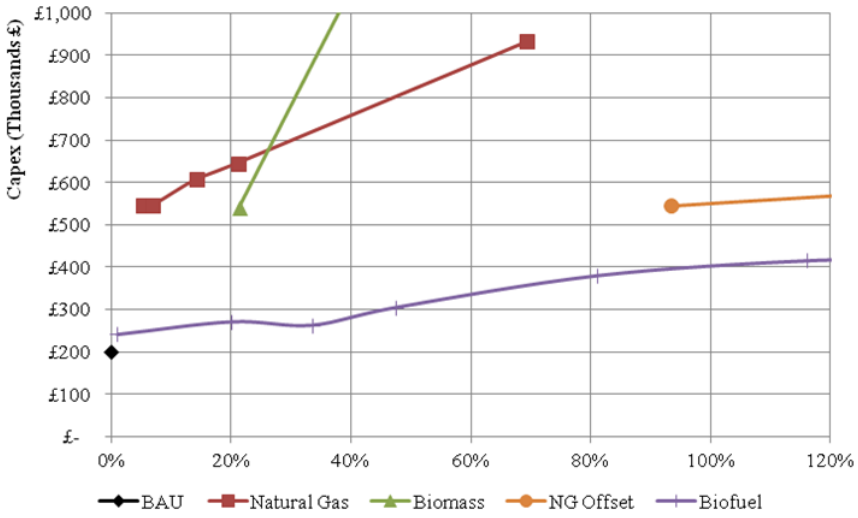

Fig. 7. Zero carbon store CAPEX results against business as usual.

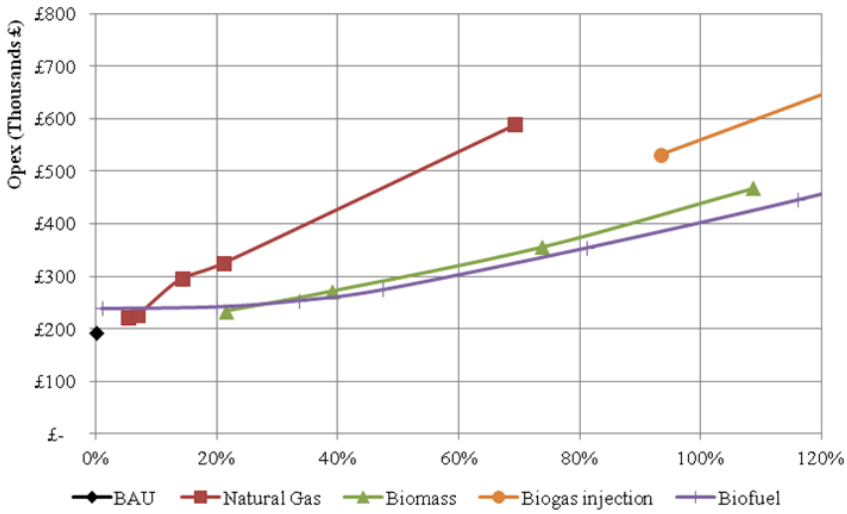

Fig. 8. Zero carbon store OPEX results against business as usual.

These two charts help greatly the decision making process. Indeed, it is showcased that the most economically viable option depends of the decarbonisation factor targeted. In terms of CAPEX, all the low carbon solutions involve a significant higher upfront cost than the BAU scenario (where the only expenditure is a conventional gas boiler). In terms of OPEX, biofuel appears as the most promising option in a large decarbonisation range. However, if the building would require more heat, then a biomass solution could be attractive. From the most interesting alternatives we can add:

- Biofuel CHP would run on Triglyceride. Fuel delivery logistics would need to be addressed as it was estimated that the site would require one fuel delivery every 5 days with a 30,000 litres tank.
- Biogas injection is probably the most pragmatic solution as it would mean that the supermarket would actually have a natural gas engine onsite connected to the grid but that the store would claim green certificates from biogas injection into the gas grid to offset its emissions. There is very little experience in the UK of such scheme and regulation is not even in place to be implemented.

As explained in the previous section, the store's decision does not rely exclusively on financial and carbon metrics, but also reducing heat waste as much as practicable should be considered. Figure 9 illustrates the percentage of heat wasted depending of the different CHP engines considered. The graph also compares the amount of heat wasted if an absorption chiller is added.

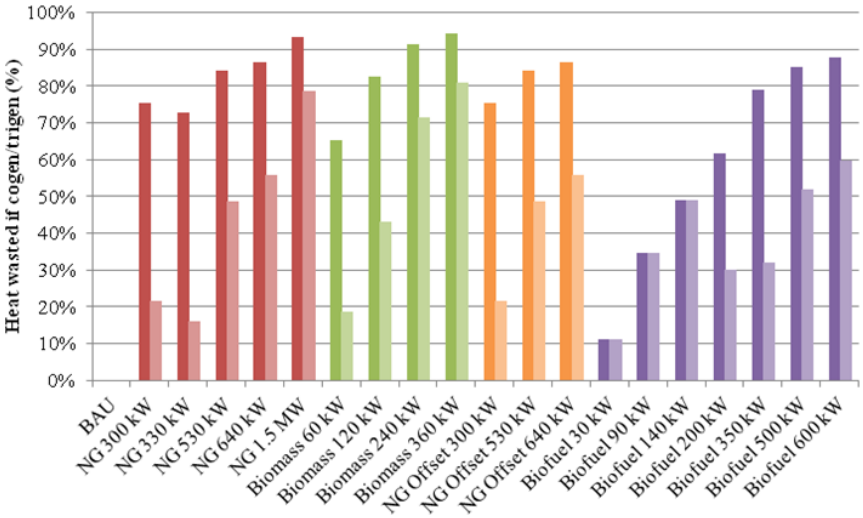

Fig. 9. Zero carbon store – percentage of heat wasted (dark = CHP, light = CCHP).

In dark color is the share of heat wasted in the context of a conventional CHP scheme, in light colour is the percentages of heat wasted if part of the wasted heat is diverted to an absorption chiller. Thus, the absorption chiller can significantly reduce the amount of heat wasted by up to 57% in the best case and in some solutions heat dumped can be lower than 20%. However, in most cases and beyond 500 kW electrical output, the percentage of heat wasted increases rapidly as the absorption chiller reaches its maximum capacity with respect to stores refrigeration demand.

5. Conclusions

Defining the technologies going into a sustainable building is just one step towards achieving a zero carbon supermarket, nonetheless it is its most critical stage, where trade-offs are assessed and priorities are given to guarantee the success of such ambitious projects. In this work the authors have focused on reviewing the present technologies that could deliver a net zero carbon building using a supermarket in the UK as a case-study.

In order to comprehend the state-of-the art of this research area the definitions of what a zero carbon building actually means were covered. Although the term can be interpreted differently, for the purpose of this paper, the authors applied the concept of a net zero carbon supermarket (NZCS). The word net adopted implies an interaction with the utility grid. The metric that is used is the GHG emissions on a yearly basis and heat and electricity use are considered for the balance. This terminology was used as it seems the most feasible to achieve based on the current technologies available.

Consequently, the text reviewed and highlighted the most appropriate low carbon technologies in a commercial environment; it was found CHP technologies are capable to reduce emissions – nevertheless the track record of these solutions is not yet solid. Meanwhile, our review found that there are other technologies like PV and wind power that unfortunately are non-dispatchable and therefore cannot necessarily offset carbon when desired. Furthermore, these technologies were not considered in the analysis as preliminary calculations showed that the capacity of PV systems on roofs is too limited and that wind turbines are not suitable in the urban environment.

An important aspect that this paper covered was describing the characteristics of heat and electricity demands in a supermarket. This in turn allowed us to perform a case-study that assessed a CCHP solution against a BAU scenario. In the case-study multiple fuel and engine sizes were evaluated with a wide range of results that confirmed the high CAPEX and OPEX these projects represent. Output shows biofuel solutions seem to be the less expensive both in CAPEX and OPEX while also having the greatest decarbonisation potential. This is mainly due to two reasons: biofuel incentives

reduce its operational costs and its modified biodiesel engine has a lower CAPEX and good power efficiencies that allows the unit to achieve high decarbonisation targets. Results show a zero carbon store is indeed feasible although it might represent challenging to make the business case work. An issue that was not explored in this paper but requires further work is how the low carbon solutions presented here compare to green power purchase agreements that off-set the carbon being emitted by buildings. The cost difference and policy in this topic could become a decisive factor in determining if net zero carbon buildings press ahead or not.

The scope of the paper is limited in defining a technology solution for a zero carbon building. But let's not forget that once a technological solution has been selected much work is still to be done. The implementation phase requires much time and effort that leads to a successful delivery. This is because independently of the solution chosen the mix and match of technologies will influence the efficiency of the whole system, however externalities will also drive the final design. Fig. 10 provides a high-level view of the many issues that need to be addressed in installing a state-of-the-art technology in a commercial environment; these are classified into:

- Smart operating systems: an effective management system needs to be in place to coordinate the interactions between the different systems present in the building; this in turn should help to improve overall system performance.
- Systems integration: careful engineering design and commissioning is fundamental so whatever equipment installed can work as intended (e.g. building management system, CHP, chillers, and refrigeration systems)
- Maintenance: proper education and awareness is to be had by the team responsible of making sure the equipment performs as expected during the life of the project.

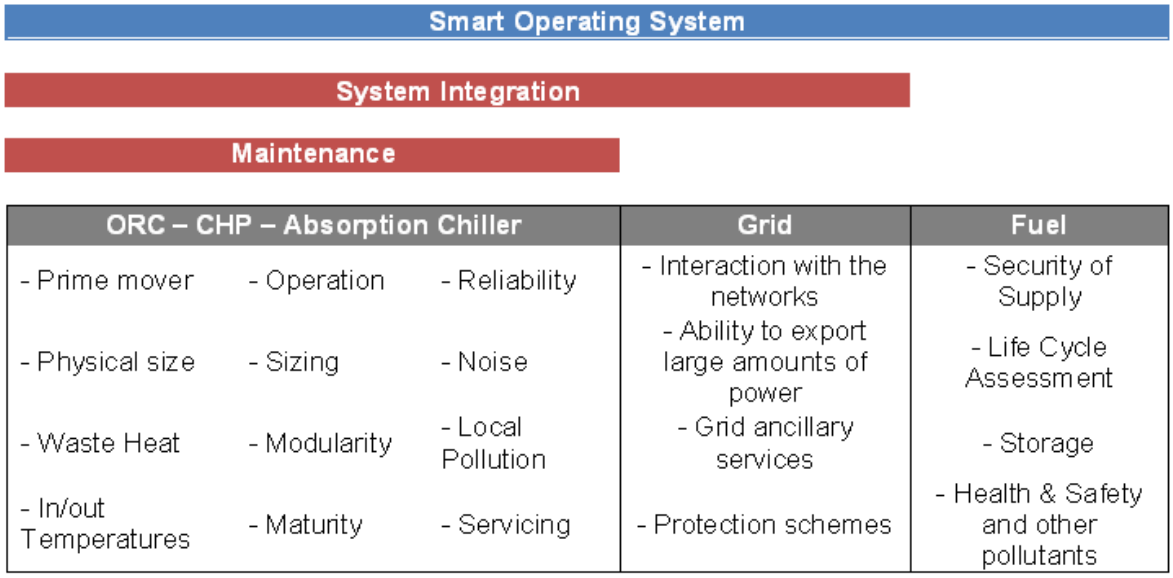

Fig. 10. Implications of delivering a net zero carbon store.

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